

Adjustment for Operation with Light load

7. Start the motor with no-load / light-load, and operates to 1/2 of the rated rotor speed
 - A1. Start operation direction:
 - a. If the start operation direction is wrong
 SPM: increase the current proportion for Pr.10-42 (Initial angle detection pulse value) to improve the accuracy of the angle detection.
 IPM: Increase the voltage for Pr.10-52 (Injection magnitude) to improve the accuracy of the angle detection.
 - b. If an ocA error occurs when pressing RUN to start the motor, decrease the current proportion for Pr.10-42 (Initial angle detection pulse value). An excessive pulse current may cause ocA error easily.
 - A2. Operates the motor in 1/2 of the rated rotor speed, adjust the no-load operating current
 If the no-load operating current exceeds 20% of the rated current, increase Pr.07-26 (Torque compensation gain) and observe the no-load operating current.
 - A3. Accelerate to rated frequency and observe if the motor operates stably.
 - a. If the motor output rotor speed presents periodic low-frequency wave, increase Pr.10-34 (PM sensorless speed estimator low-pass filter gain), or increase Pr.10-32 (PM FOC sensorless speed estimator bandwidth).
 - b. If the output frequency reflects high frequency vibration, decrease Pr.10-34 or decrease Pr.10-32.
 - A4. Accelerate the motor to the maximum rotor speed and observe if it operates stably.
 If the motor stalls when accelerating to the maximum rotor speed, then increase Pr.10-34 PM Sensorless Speed Estimator Low-pass Filter Gain, or increase Pr.00-17 Carrier Frequency (you must set the carrier frequency larger than 10 times of the maximum output frequency)